

KIRBYMIMI PRESENTS



ADVANCED SETUP



# SUMMARY

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- How To Setup Client Mod
- How To Setup the Server & Guides
  - Regular Hosting Setup & Guide - ZonaiHosting Setup & Guide
  - Docker Setup & Guide - Zonai Realms Hosting Setup & Guide
- Save Editing
- Network
- Early Stages & Bugs



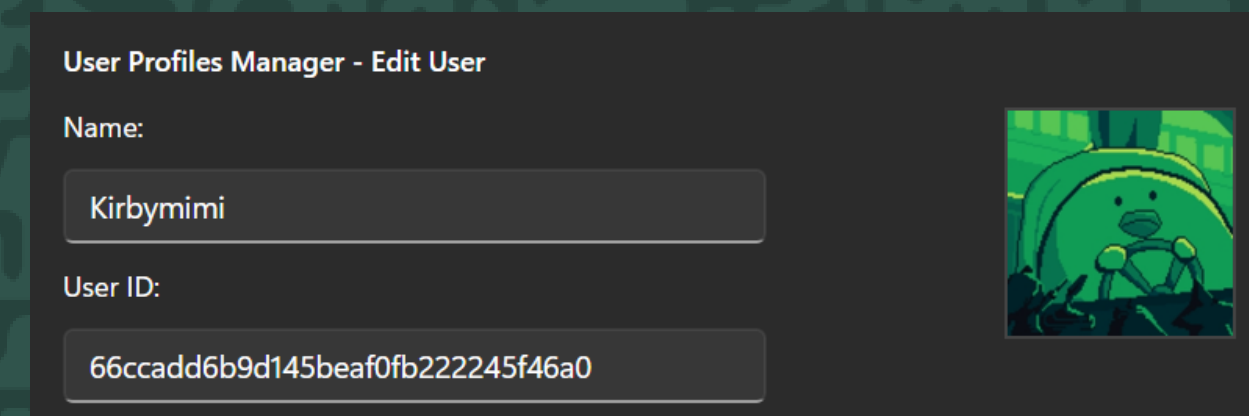
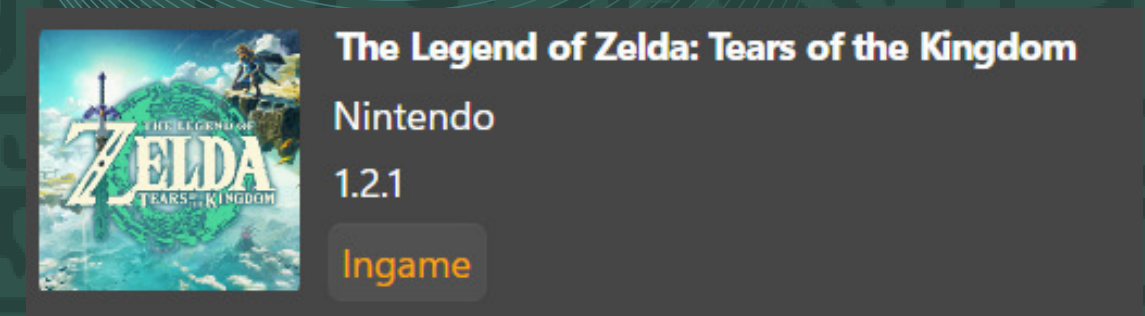
**Versions of the mod can be found here:**  
<https://discord.com/invite/ZPvZm3NFak>

**Support Kirbymimi and future projects here:**  
<https://patreon.com/kirbymimi>

# Pre-Mod Preparations

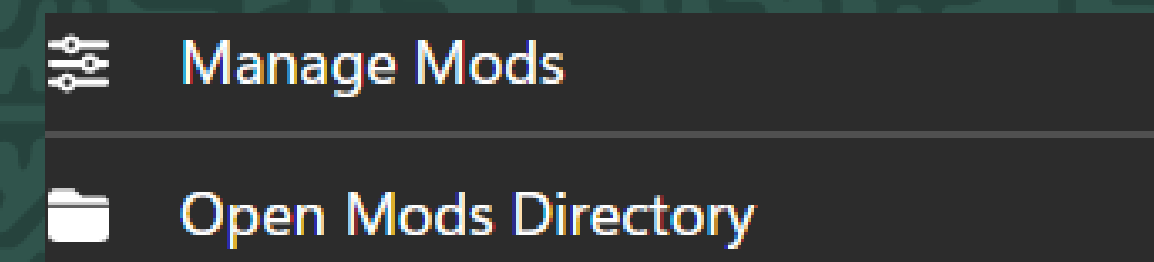
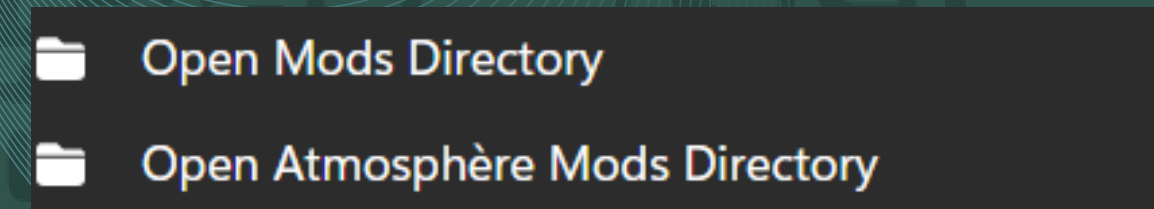
**This tutorial will be mainly done in Ryujinx, the recommended emulator, or the latest fork Ryubing. For other emulators or systems, find tutorials online or ask for help on the Discord.**

- Be sure you are on Tears of the Kingdom Update 1.2.1.
- In order to be able to play, you must have a pre-existing save on the game, clicking new save will not work. Autosaves will be overwritten with different ones, so backup your saves prior to playing. These are not actual server saves and are saved on the servers end. If you're on Switch, be sure your playing with an account.
- On emulator, make a save profile with a name and picture, you cannot play the mod without a form of unique user ID.



# How To Setup Client Mod

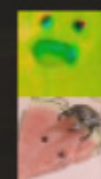
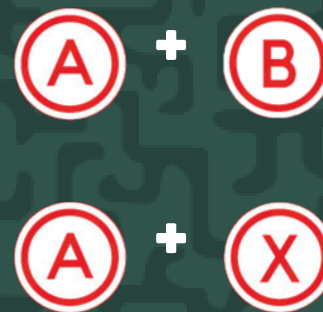
- This depends on the platform you're using, but the mods directory will typically be under the games title ID. For Ryujinx, open the application and right-click Tears of the Kingdom, then click "Open Mods Directory".
- Open the mod files and get the client mod folder that doesn't have "server" in it. Take that file and put it in the Mods Directory.
- If the game doesn't enable the mod by default, right click the game again and click "Manage Mods". Then, simply enable the mod.



# Game Information

## Using the chat

The BasicUIs plugin adds a chat that all connected users can use, simply hold "A+B" for a few seconds to show/hide it, and hold "A+X" to open a keyboard to send a message.



mimi :3: This is a chat  
not mimi: I am schizophrenic

# How To Setup the Server & Guides

This part varies depending on how you want to host. There are many intricate advantages, downsides, and other features provided by each form of setup.



For everything you will need Java 25.

found here:

<https://www.oracle.com/java/technologies/downloads/#java25>

This list will go from least intricate to most intricate in the following order:

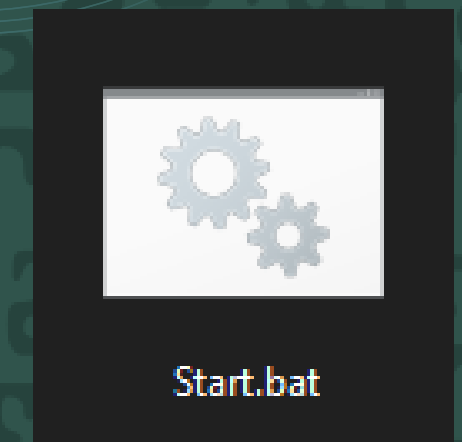
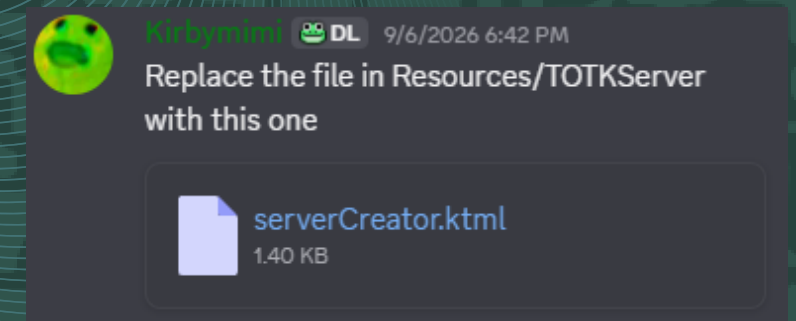
1. Regular Hosting Setup & Guide
2. ZonaiHosting Setup & Guide
3. Docker Setup & Guide
4. Zonai Realms Hosting Setup & Guide

Refer to Regular Hosting Guide to understand basic ideas for server management.

# Regular Hosting Setup

As of right now, in order to run the server regularly, you have to modify the upload speed of the server to a perfect medium for everyone wanting to play, or they will load into the game infinitely or sit on a black screen. This can be fixed by copying the file pinned in Kirbymimi's Discord TOTK channel and following the path Kirby put. If the speed doesn't work, modify the "400000" variable in the file to your own. Lowering the speed will make other player movements look jittery as if they were teleporting, so use another method if this is an issue for you.

- Simply open the mod files and open the server folder.
- click and open "start.bat".
- The server should run on default parameters with no JVMCI/GraalVM (These are utilized in Docker Setup.)



```
WARNING: A restricted method in java.lang.System has been called
WARNING: java.lang.System::load has been called by com.oracle.truffle.polyglot.IDCSupport in an unnamed module (file:/C:/Users/yashy/Desktop/Root/Epic/EclipseWorkspace/totkServer/release)
WARNING: Use --enable-native-access=ALL-UNNAMED to avoid a warning for callers in this module
WARNING: Restricted methods will be blocked in a future release unless native access is enabled

WARNING: A terminally deprecated method in sun.misc.Unsafe has been called
WARNING: sun.misc.Unsafe::objectFieldOffset has been called by com.oracle.truffle.api.strings.TStringUnsafe (file:/C:/Users/yashy/Desktop/Root/Epic/EclipseWorkspace/totkServer/release/)
WARNING: Please consider reporting this to the maintainers of class com.oracle.truffle.api.strings.TStringUnsafe
WARNING: sun.misc.Unsafe::objectFieldOffset will be removed in a future release
[To redirect Truffle log output to a file use one of the following options:
* '--log.file<path>' if the option is passed using a guest language launcher.
* '-Dpolyglot.log.file<path>' if the option is passed using the host Java launcher.
* Configure logging using the polyglot embedding API.]
(engine) WARNING: The polyglot engine uses a fallback runtime that does not support runtime compilation to native code.
Execution without runtime compilation will negatively impact the guest application performance.
The following cause was found: JVMCI is not enabled for this JVM. Enable JVMCI using -XX:+EnableJVMCI.
For more information see: https://www.graalvm.org/latest/reference-manual/embed-languages/trustee-optimization-support.
To disable this warning use the '--engine.WarnInterpreterOnly=false' option or the '-Dpolyglot.engine.WarnInterpreterOnly=false' system property.
Server initialized successfully
```

# Regular Hosting Guide

Most of the base-line server management can be split in 5 groups:



## Save Data

Save data on the server is split into 3:

- User/SaveServer/Save.html (Global Progression Save, including Shrines, Korok Seeds, Light of Blessing, Abilities, etc.)
- User/SaveServer/users (Playersaves including Hearts, Inventory, Stamina, etc.)
- Resources/SaveServer/defaultClientSave.html (Starting playersave for people just joining.)

Global saves can be imported by putting a progress.sav into the same folder as the global save. For playersaves refer to the Save Editing Section.

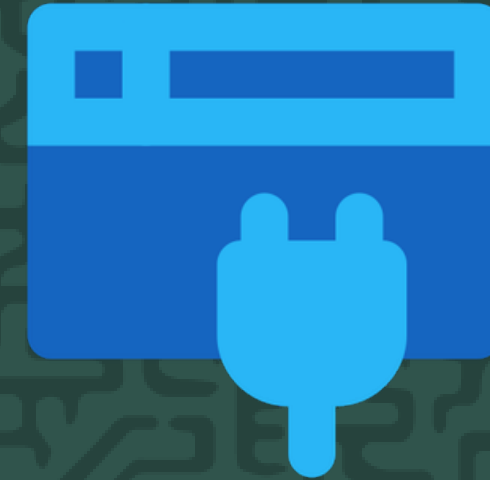
Global and player saves are fundamentally different.



## PuppetPacks

PuppetPacks folder holds pvp and collision capabilities that can be put to replace the PlayerPuppet.pack.zs in Romfs/Pack/Actor.

- Dummy has no pvp and collision.
- Physics has pvp and collision.



## Plugins

Plugins written in Python can be loaded by the server and executed by the server and the client, the server ships with the "BasicUIs" plugin inside the "Plugins" folder, you can read and modify it to learn how to write new plugins, all functions callable by the client can be found in "PythonAPI/Client".



## Romfs

Romfs are a group of mods that can be put on serverside, for all players to experience.

Exefs are currently unable to be put on the server, so if a mod uses the exefs folder, you will need to put the exefs folder on client-side. Think of it like a client-side Minecraft mod you want on a server, everyone must download it to experience the same thing.

Mods that use the same files modified in the servers Romfs folder must be merged using Simple Mod Manager (SMM) or TKMM, Be sure to backup the original folder incase of compatibility issues.



## Ports

Port can be modified in Resources/TOTKServer/serverCreator.html

Server port is 10014  
Localhost would be 127.0.0.1:10014

# ZonaiHosting Setup

If infinite loading is still an issue on this (very rarely), then check Upload speed on the ZonaiHosting Guide next.

- Head over to:  
<https://zonaihosting.lonlonlabs.dev/>
- click "Open Room" with however much time you want (max 8 hours before server stops.)
- Enjoy Playing with your friends on max-capability servers! You can also use console commands; a list can be found by typing "help".

Check out Wesley's amazing projects here:

<https://discord.gg/VeccXh4ydN>

## Zonai Hosting

Temporary Tears of the Kingdom Online rooms for playing with your friends the easy way.

 Sounds: On

### Open a room

Open for how long? (up to 8 hours)

Open room

Up to 2 rooms per network. A random port between 10000 and 12000 is assigned when it opens.

# ZonaiHosting Guide

- For the server's safety, turn your server off before modifying anything on this list. Port is NOT manageable on these servers, and provided to you.

### Custom save

Stop the room first. Uploading replaces the current save with your .sav file.

Choose File

No file chosen

Upload save

Download current server save as 

.ktml

Download

.sav conversion uses the game server's own save format exactly (ported directly from its save-conversion code), separate converter available (created by [ThyHeroOfTime](#)).

## Custom save

- Global saves can be modified and downloaded from the Server save as a .ktml. Conversion can be found in the Save Editing Section

### Default client save

Resources/SaveServer/defaultClientSave.ktml for this room -- the template every new player who joins gets seeded from. This only affects players who haven't connected to this room yet.

Download

Choose File

No file chosen

Upload & replace

Stop the room first to upload.

- Default player save (or defaultClientSave.ktml) can be modified and downloaded in this section.

### Players

|   | Name   | Last seen                      |                                                                                              |
|---|--------|--------------------------------|----------------------------------------------------------------------------------------------|
|   |        |                                | Download save (.ktml)                                                                        |
| • | Wesley | 2026-09-09T04:58:36.270137 UTC | <div><div>Browse...</div><div>N....</div><div>Upload save</div></div> <div>Delete save</div> |
|   |        |                                | Download save (.ktml)                                                                        |
| • | Wesley | 2026-09-08T21:17:26.900419 UTC | <div><div>Browse...</div><div>N....</div><div>Upload save</div></div> <div>Delete save</div> |

- Playersaves can be modified in this list. depending on if the server is on or off, you will have more options when it is off to edit/upload, download, or delete saves.

### Mods (Romfs)

Stop the room first. Upload a .zip of your Romfs mod files — matching files are overwritten, new files are added, and anything else already there is left alone. Zip either your Romfs folder itself, or just its contents (Pack/, System/, etc. at the top level).

Choose File

No file chosen

Upload mods

## Mods (Romfs)

- Self explanatory. Use TKMM or Simple Mod Manager to merge mods together with the Server Romfs file found in Kirby's release on his Discord, then upload the Romfs folder here.

### PVP

Stop the room first. Controls whether other players' puppets can collide with and hit you.

• PVP is currently off

Enable PVP

## PVP

- Swaps out the PuppetPacks on its own to enable or disable pvp physics on your server.

### Upload speed

Stop the room first. Caps how fast players can upload data (mods, saves, etc.) to the server, in bytes/second. Default is 10,000,000 (10 MB/s) — if someone is having trouble connecting, try lowering this.

Currently 10,000,000 bytes/second

10000000

Save

## Upload speed

- Changes the upload speed used in Resources/TOTKServer/serverCreator.ktml. If you are experiencing infinite loading or blackscreen, try lowering this.

# Docker Setup

Docker is an incredibly technical setup that requires a bit of knowledge about running terminals/command prompts. This will get full capability of your server in terms of speed and optimization.

- Download the Docker application here: <https://docs.docker.com/desktop>

- On Windows, you will also need to download WSL (Windows Subsystem for Linux) here: <https://github.com/microsoft/WSL/releases/>

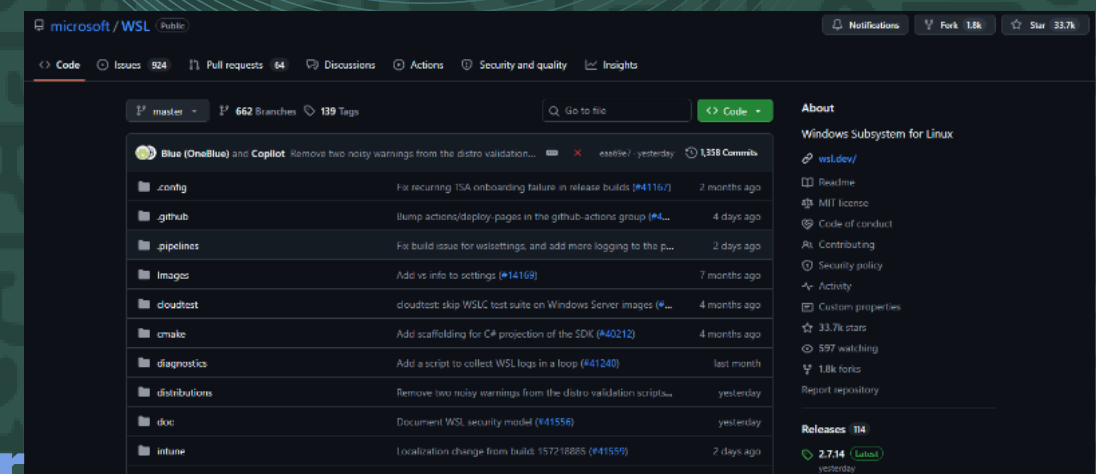
- After setting up Docker, go to [https://hub.docker.com/r/wesleyhellewell/totk\\_online\\_multiplayer\\_server](https://hub.docker.com/r/wesleyhellewell/totk_online_multiplayer_server) (vanilla Docker server) or [https://hub.docker.com/r/wesleyhellewell/totk\\_online\\_multiplayer\\_server\\_test](https://hub.docker.com/r/wesleyhellewell/totk_online_multiplayer_server_test) (Docker with Wesley's patches and fixes)

- Copy the "docker pull" line on the "Overview" page.

- In a new folder in your files, make a file called "docker-compose.yml", then edit that file in Notepad and put the "docker pull" line you copied, then hit save.

- Open terminal or command prompt in that folder ("cmd" in file path for windows) and type/run "docker compose up -d".

- Alternatively, you can make a start.bat file that contains "docker compose up -d". both methods will create all the server files for you on your computer.

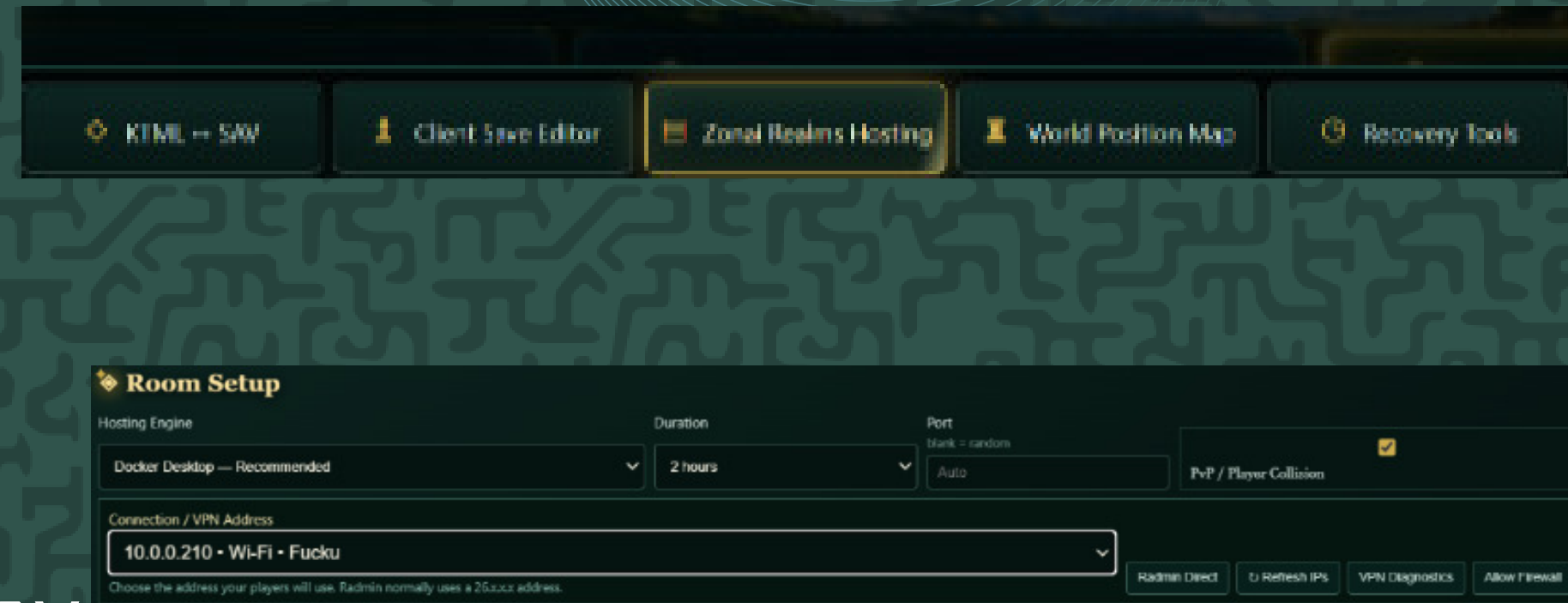


# Docker Guide

- Use Regular Hosting Guide for Reference, as all files are located in the folder with docker-compose.yml.
- Docker console is in docker itself under its corresponding container in the "Containers" tab.
- You can use Wesley's console commands under this console, type "help" for a list of commands.
- Port is managed in "View configurations" by replacing 10014:10014 with NEW\_PORT:10014.

# Zonai Realms Hosting Setup

- Download this GitHub here:  
<https://github.com/LegendarySavageGamer/TOTK-1.2.1-Save-Converter/releases/>
- Download Docker (Refer to Docker Setup Section to do this)
- Launch the .exe and go to "Zonai Realms Hosting".
- Use whatever parameters you need including a Docker setup (thanks to wesley or native Java setup, uptime, port, and network/vpn (good for managing radmin and native connection))



**Check out LegendarySavageGamer's amazing projects here:**

<https://github.com/LegendarySavageGamer>

# Zonai Realms Hosting Guide

- For the server's safety, turn your server off before modifying anything on this list.



The screenshot shows the 'Connection Rune' interface. At the top, it displays the IP address '10.0.0.210:11590'. Below this is a 'Copy Connection' button. The interface is divided into two columns. The left column contains 'PORT 11590' and 'ENGINE DOCKER'. The right column contains 'TIME 00:00:00' and 'PLAYERS 0'. Below these are four buttons: 'Open Realm Folder', 'Refresh Docker Java', 'Server KTML', and 'Server SAV'. At the bottom, there are two more buttons: 'Upload Main Save' and 'Merge Romfs ZIP'. A status bar at the very bottom shows 'Realm folder opened.' with a small yellow star icon on the right.

## Connection Rune

- Displays server stats, file save location via "Open Realm Folder", Refresh the Docker, "Server KTML" and "Server SAV" to download the global save. "Upload Main Save" to upload a global save. "Merge Romfs ZIP" to upload mods (pre-merge with server Romfs files provided in Kirby's Discord for safety using TKMM or Simple Mod Manager), etc.



The screenshot shows the 'Maximum Resource Upload' interface. It features a slider control for setting the upload speed. The current value is '10000000'. To the right of the slider are four buttons: '1 MB/s SAFE', '2.5 MB/s', '5 MB/s', and '10 MB/s ORIGINAL'. Below the slider, there is a warning message: 'Black-screen warning: 1 MB/s is the tested safe setting. Raise this only if every player can connect reliably.'

## Maximum Resource Upload

- Changes the upload speed used in Resources/TOTKServer/serverCreator.ktml. If you are experiencing infinite loading or blackscreen, try lowering this.



The screenshot shows the 'Travelers in the Realm' interface. It has a title bar with a yellow star icon and the text 'Travelers in the Realm'. Below the title bar, there is a small text area that says 'No player saves mapped yet. Players will appear after they connect.' There is a 'Refresh Now' button in the top right corner.

## Travelers in the Realm

- Lists and modifies players and their playersaves.

## Room Setup

Port and Connection/VPN (selecting the connection you use for IP) can be changed here.



The screenshot shows the 'Room Setup' interface. It has a title bar with a yellow star icon and the text 'Room Setup'. Below the title bar, there are three dropdown menus: 'Hosting Engine' (set to 'Docker Desktop - Recommended'), 'Duration' (set to '2 hours'), and 'Port' (set to 'Auto'). To the right of the 'Port' dropdown is a checkbox labeled 'PvP / Player Collision' which is checked. Below these is a dropdown menu for 'Connection / VPN Address' set to '10.0.0.210 - Wi-Fi - Fucku'. At the bottom, there is a small text area that says 'Choose the address your players will use. Radmin normally uses a 255.255.255 address.' and four buttons: 'Radmin Direct', 'Refresh IPs', 'VPN Diagnostics', and 'Allow Firewall'.

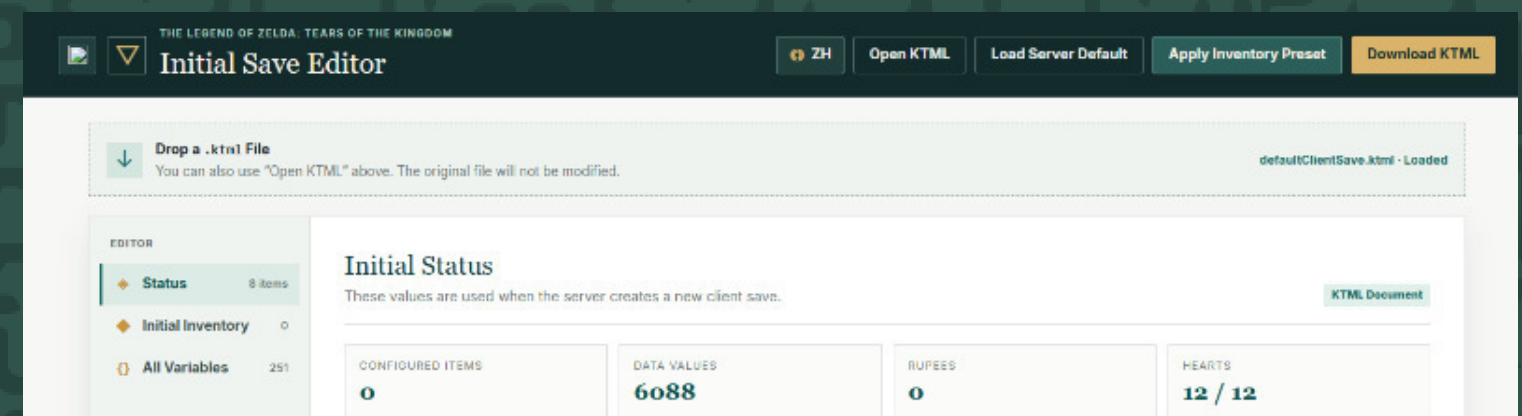
# Save Editing

There are currently a couple of ways to edit saves or other info on your server:

- ZonaiRealms which is basically an all-in-one server editor with ktml-sav-ktml conversion, client (or player) save editors, positions selection map (good for recovery from the void), and much more in future updates:

<https://github.com/LegendarySavageGamer/TOTK-1.2.1-Save-Converter/releases/>

- Tiang-88's defaultClientSave editor, which allows you to modify playersaves and defaultclientsaves with any inventory items and player values: <https://github.com/Tiang-88/zelda-totk-Online-Mod-defaultClientSave-editor>



# Network

- For regular hosting, you will need the hosts public IPv4 and to be able to port forward, which sends network traffic for your port to your server. Find your public IP here if needed (or in your network settings): <https://www.whatismyip.com/>
- Third party methods can be used if you are unable to port forward, like playit.gg which tunnels specific network connections to your selected server. Then there is also things like Zonai Hosting, which do it all on their end.

# Early Stages & Bugs

Not much is really know about the mod in its early stages, so there arent many real solutions to "Switch crashing or blackscreened" or "Game crashes after 5 minutes" or anything mod related.

For the best support on this, ask on Kirbymimi's discord through the TOTK channel or the assistance forum. If you experience crashes, please send any crash logs from you emulator or server console. A good method to test for bugs would be to try on other emulators.

Do not directly mention Kirbymimi, he works hard to bring his members and patrons quality mods that take time.

